

LVDS Transmitter with Integer/Fractional PLL

I/O Datasheet

Chipalooza 2026 · IHP SG13CMOS5L

Revision of 1 September 2026

1 Description

A PRBS-7 pattern generator and an LVDS output driver, clocked either directly from the reference or from an on-chip PLL whose feedback ratio is programmable over the shared digital inputs. The port list is `verilog/rtl/user_project_wrapper_4a.v` of `RTimothyEdwards/sg13cmos5l_ocd_chipalooza`, with every bus expanded into individual pins. Four dedicated analog pads means wrapper `_4a`, and therefore slot `s1` or `s16` — per `config.txt` the only two with four.

Signal flow: the reference arrives on `clk`; the PLL multiplies it by `DIV_RATIO` and halves the result; `lvds_pattern` selects reference or PLL clock, generates PRBS-7 and hands a differential pair to `lvds_tx`, which drives two of the analog pads.

2 Pin summary

All 54 ports of the wrapper. *Not connected* means the pin exists in the port list but reaches no logic inside this project.

Pin	Dir	Function
<i>Supplies</i>		
<code>vdd_3v3</code>	inout	3.3 V — driver, pre-driver, output stage
<code>vss_3v3</code>	inout	3.3 V return
<code>vdd_1v2</code>	inout	1.2 V — pattern generator, PLL
<code>vss_1v2</code>	inout	1.2 V return
<code>vssio</code>	inout	not connected
<i>Clock and enable</i>		
<code>clk</code>	in	PLL reference, 25 to 250 MHz verified
<code>enable</code>	in	not connected — see §6
<i>Pattern generator control</i>		
<code>dig_in[0]</code>	in	<code>clk_src</code> : 0 takes <code>clk</code> , 1 takes the PLL
<code>dig_in[1]</code>	in	<code>en</code> — gates the pattern clock
<code>dig_in[2]</code>	in	<code>reset</code> , active high, seeds the PRBS
<code>dig_in[3]</code>	in	<code>mode</code> : 0 clock passthrough, 1 PRBS-7
<code>dig_in[4]</code>	in	not connected
<i>PLL control</i>		
<code>dig_in[5]</code>	in	PLL ENABLE
<code>dig_in[6]</code>	in	PLL RESET_N, active low
<code>dig_in[16:7]</code>	in	<code>DIV_RATIO[9:0]</code> , unsigned Q7.3 — see §4
<code>dig_in[18:17]</code>	in	<code>TEST_DIV[1:0]</code> — selects the <code>TEST_CLK</code> divider
<code>dig_in[23:19]</code>	in	not connected
<i>Digital outputs</i>		
<code>dig_out[11:0]</code>	out	not connected
<i>Dedicated analog pads</i>		
<code>analog_pin[0]</code>	inout	spare
<code>analog_pin[1]</code>	inout	<code>TEST_CLK</code> brought off chip
<code>analog_pin[2]</code>	inout	LVDS output, non-inverting
<code>analog_pin[3]</code>	inout	LVDS output, inverting

Pin	Dir	Function
<i>Bias references</i>		
ibias[0]	in	2 μ A — pre-driver, mirrored 1:15 internally
ibias[1]	in	2 μ A — driver, mirrored 1:15 internally
vbias	in	not connected
analog_bus[0]	inout	2 μ A — PLL charge pump
analog_bus[1]	inout	1.2 V — LVDS common-mode reference
analog_bus[3:2]	inout	not connected

3 Recommended operating conditions

Parameter	Pin	Min	Typ	Max	Unit
Core supply	vdd_1v2	1.08	1.20	1.32	V
Driver supply	vdd_3v3	2.97	3.30	3.63	V
Reference clock	clk	25	250	500	MHz
Bias current	ibias[1:0]	0.05	2	10	μ A
Charge-pump bias	analog_bus[0]	0.05	2	10	μ A
Common-mode ref.	analog_bus[1]	–	1.2	–	V

The bias limits are those of the harness current DACs, whose range is 50 nA to 10 μ A. The blocks themselves were characterised at 30 μ A; the 1:15 pre-mirrors in `lvds_tx` bridge the gap, so the pins take 2 μ A while the driver and pre-driver still see the bias they were measured at.

4 PLL configuration

The feedback ratio is one unsigned Q7.3 word on `dig_in[16:7]`: `div_ratio[9:3]` is the integer part and `div_ratio[2:0]` the eighths. `fractional_divider.v` requires the integer part to be at least 4, so the usable range is 4.000 to 127.875 in steps of 0.125.

$$f_{VCO} = f_{clk} \times \text{DIV_RATIO} \quad f_{pll_clk} = \frac{f_{VCO}}{2} \quad f_{TEST_CLK} = \frac{f_{VCO}}{2^{1+\text{TEST_DIV}}}$$

The output divider always halves, so the line rate equals $f_{clk} \times \text{DIV_RATIO}/2$. The ring oscillator covers 322 to 4050 MHz at typical / 1.2 V / 27 °C and does not start below 0.50 V of control voltage, so combinations whose VCO frequency falls outside that band will not lock.

Worked example. 250 MHz on `clk` and a line rate of 500 Mbit/s wants `DIV_RATIO` = 4.0, code $4.0 \times 8 = 32 = 0000100000_2$. Only bit 5 is set, which is `dig_in[12]`; every other ratio bit is driven low. The VCO then runs at 1 GHz.

Fractional operation. The divider is a plain $N/N+1$ accumulator with no delta-sigma shaping. At a fractional ratio the instantaneous period alternates and only the average over a full accumulator cycle is the intended frequency; expect corresponding spurs on the control voltage.

5 Electrical characteristics

Simulated at the top level, typical corner, 27 °C, over 6 μ s, with the reference at 250 MHz and `DIV_RATIO` = 4.0. Load is 49.9 Ω + 49.9 Ω across the two output pads with the centre tap giving V_{OS} . **No pad or ESD model is included**, so V_{OS} here is not the compliance figure.

Parameter	Symbol	Measured	Limit	Reference
Differential output	$ V_{OD} $	380.5	247 ... 454	TIA/EIA-644-A 4.1.1
Offset voltage	V_{OS}	1193.8	1125 ... 1375	4.1.2
Offset, peak-to-peak	$V_{OS,pp}$	120.4	< 150	4.1.5
PLL output frequency	f_{pll_clk}	500.0038	500	target
Frequency error		7.5		ppm
Control voltage	V_{CTRL}	581.4		mV, settled
Driver supply current	I_{3v3}	5.74		mA
Core supply current	I_{1v2}	0.311		mA

Voltages in mV unless noted. The frequency and control-voltage figures come from the co-simulated PLL, in which the phase detector and both dividers are the RTL of `macros/pll_digital` executed through ngspice's `d_cosim`. Supply currents are from the LVDS bench, which clocks the pattern generator from the reference directly.

6 Known limitations

- **No ESD structures or pad models.** Clause 4.1.4 and 4.1.5 figures depend on pad capacitance, which is not represented here.
- **enable has no load.** The pattern clock is gated by `dig_in[1]` alone. The harness masks `dig_in` to zero for an unselected project, so an unselected project is still idle — but `select` and `dig_ena` high with `enable` low leaves the block running with the project enable deasserted.
- **ΔV_{OS} is not measured at the top level.** The 50 mV limit of clause 4.1.2 is checked only in the macro's own DC bench.
- **Offset margin is asymmetric.** At the 1.2 V reference — the nearest value on the 0.3 V adjustment grid — V_{OS} sits 69 mV above the lower limit and 181 mV below the upper. Corner coverage is outstanding.
- `analog_pin[1]` is driven by `TEST_CLK` with no pad driver in front of it.
- `core_p` / `core_n` cross from `vss_1v2` into the pre-driver on `vss_3v3`; the two returns have to be tied in the frame.
- **Simulation needs a build step.** The PLL's digital half runs as a Verilator-compiled shared object under `d_cosim`. Run `make pll-cosim-so` before any top-level simulation; the Makefile and the Simulate arrow in xschem both do this automatically.